

1 HIV|SIR Model

HIV: $\begin{cases} \dot{x} = \lambda - dx - \beta xv \\ \dot{y} = \beta xv - ay \\ \dot{v} = ky - uv \end{cases} \Rightarrow \begin{cases} \lambda: \text{formation rate of healthy cells.} \\ d: \text{death rate of healthy cells.} \\ \beta: \text{infection rate.} \\ a: \text{death rate of infected cells.} \\ k: \text{formation rate of free viruses.} \\ u: \text{death rate of free viruses.} \end{cases}$

$\hat{v}$ 中缺 $-\beta xv$ 的原因: The model neglects the cost of virus when it infects healthy cells.

virus-free steady-state: $y = 0$ and $v = 0$, possible $x = \frac{\lambda}{d}$.

Estimate virus increase: in each virus generation/over the characteristic lifetime:

Have: $(x, y, v)(0) = (\frac{\lambda}{d}, 0, v_0)$. 1. Period of free virus lifetime: u^{-1} . Period of infected cell lifetime: a^{-1} .

2. $\frac{1}{u} \ll \frac{1}{a}$ 3. $v(\frac{1}{u}) - v(0) = \int_0^{\frac{1}{u}} \dot{v} dt = k \int_0^{\frac{1}{u}} y dt - u \int_0^{\frac{1}{u}} v dt$
a1. $\int_0^{\frac{1}{u}} y dt \approx \int_0^{\frac{1}{u}} y(\frac{1}{a}) dt$ b. $\int_0^{\frac{1}{u}} v dt \approx \int_0^{\frac{1}{u}} v(0) dt$, a2. $y(\frac{1}{a}) = \int_0^{\frac{1}{a}} \dot{y} dt \approx \int_0^{\frac{1}{a}} \dot{y}(0) dt$

Basic Reproductive ratio: $R_0 = \frac{k\beta\lambda}{a d u}$, $R_0 > 1$ virus invades; $R_0 < 1$ virus dies out.

估计 (e.g. R_0) 某个值范围 | 最后趋近于某个值 | 得到趋近速率: 1. 计算 equilibrium. 2. 计算 Jacobian 算 eigenvalues. 3. 取最大 eigenvalue $\lambda \rightarrow$ 得 sol $\propto e^{\lambda t}$. 4. 根据题意估计求值范围 | 趋近值 | 速率 (λ).

(e.g.) 证明 $u = \text{cont. when } t = \mathcal{O}(u^{-1})$: 1. Linearise 估算趋近值 2. 假设原式 $\approx$ 估计值 3. 解出 t 估计

SIR: $\begin{cases} S' = -\beta SI \\ I' = \beta SI - \alpha I \\ R' = \alpha I \end{cases} \Rightarrow \begin{cases} S: \text{susceptible individuals (易感人群).} \\ I: \text{infected individuals (感染者).} \\ R: \text{recovered individuals (康复者).} \\ \beta: \text{infection rate.} \\ \alpha: \text{recovery rate. (or death, if no R)} \end{cases}$

ps: $\lim_{t \rightarrow \infty} I(t) = 0$ $\lim_{t \rightarrow \infty} S(t) > 0$ and $R(0) = 0$

Final Size Relation: $\ln(S_0/S_\infty) = \frac{\beta}{\alpha} [N - S_\infty] = R_0(1 - \frac{S_\infty}{S_0})$

proof: 1. $(S + I)' = -\alpha I$ 2. $\int_0^\infty (S + I)' dt = -\alpha \int_0^\infty I dt \Rightarrow (S + I)_\infty - (S + I)_0 = -\alpha \int_0^\infty I dt$

3. By $N = S_0 + I_0$, can get $S_\infty - N = \ln(S_\infty/S_0) \frac{\beta}{\alpha}$. $\square$

2 Relaxation|Excitation|Switch - FitzHugh Nagumo Model

Explain why oscillations/relaxation occur:

1. (e.g.) $\dot{x} = c[f(x) - y]$ and $\dot{y} = a$, where $f(x)$ is a cubic-like curve. $c \gg 1$
2. show $|\dot{x}| \gg |\dot{y}|$ (or vice versa), and no stable equilibrium exists. (can unstable)
3. 图上标一起始点 a, 画类似右侧图. 4. (e.g.) a. Start at a, $|\dot{x}| \gg |\dot{y}|$, so it quickly moves horizontally to the right branch of cubic-like curve (A). b. When it closed to the right branch, since $\dot{y} < 0$, $\dot{x} \approx 0$, so it moves downwards along the right branch. Then, it closes to the point B. It has option to leave the path $f(x)$ since $\dot{y} < 0$, so it cannot follow $f(x)$ and again quickly jumps from B to the left branch of the cubic-like curve (C) since $|\dot{x}| \gg |\dot{y}|$. When it closed to the left branch, since $\dot{y} > 0$, $\dot{x} \approx 0$, so it moves upwards along the left branch. Finally, it reaches the point D, and again quickly jumps to the right branch of the cubic-like curve (A). This process repeats, resulting in relaxation oscillations.

Estimate Period: (neglect fast jumps) 1. Period $T \approx \int_t^B \frac{dy}{dx} dt + \int_C^t \frac{dy}{dx} dt = 2 \int_t^B \frac{dy}{dx} dt$ (by symmetry)

2. (e.g.) $\dot{x} = c[f(x) - y]$, $\dot{y} = g(x, y)$ then $y \approx f(x)$, $\frac{dy}{dx} = \frac{dy}{dx} \frac{dx}{dt} = f'(x) \frac{dx}{dt} \Rightarrow dt = \frac{f'(x)}{g(x, y)} dx$

3. Find coordinates of A, B, C, D (turning points) $\Rightarrow T \approx 2 \int_A^B \frac{dx}{f(x) - y}$

Three kinds of Excitable System:

small perturbation: returns to equilibrium. (fail excitation)

large perturbation: large excursion in phase space before returning to equilibrium. (action potential|successful excitation)

Refractory period(不应期): system insensitive to further

constant stimulus: periodic excitation. (repetitive firing|limit cycle)

Explain why excitable occur: 结合前面 relaxation oscillation 的分析,

stable equilibrium (图为原点), 分析 P|A 开始 (small|large perturbations).

FitzHugh-Nagumo Model:

Have: $\begin{cases} \dot{v} = f(v) - w + I(t) \\ \dot{w} = bv - \gamma w \end{cases} \Rightarrow \begin{cases} v: \text{membrane (excitable) potential.} \\ w: \text{ionic current. (recovery var)} \\ a, b, c: \text{positive constants.} \\ I(t): \text{external stimulus.} \end{cases}$

Switch Behaviour: 类似 relaxation oscillation, 有两个 stable equilibria, 分析一个过去后不会回来.

Remark: 对于这类模型, 计算平衡点用 $f(v)$ 的图像斜率和另一条直线斜率来判断.

3 Stability|Linearise Stability

Continuous $\frac{dN}{dt} = f(N)$:

- **Stable:** $f'(N^*) < 0$

- **Unstable:** $f'(N^*) > 0$

Discrete $N_{t+1} = f(N_t)$: $\lambda = f'(N^*)$

- **Stable:** $|\lambda| < 1$ - **periodic:** $|\lambda| = 1$

- **Unstable:** $|\lambda| > 1$ (chaotic)

- **tangent bifurcation:** $\lambda = 1$

- **period-doubling bifurcation:** $\lambda = -1$

One var. $\uparrow$ | More var. $\downarrow$:

Jacobian: $J = [F_x \ F_y \ \dots]$

Eigenvalue Method:

If $\text{Re}(\lambda_i) < 0$ for all i , then asy stable.

If $\lambda_1 < 0 < \lambda_2$, then saddle (unstable).

If $\text{Im}(\lambda_i) \neq 0$, then focus (spiral).

If $\text{Re}(\lambda_i) = 0$, then center (linear!).

or (nonlinear): Hopf bifurcation.

If $\lambda = 0$: If $\lambda_1 = 0$, $\lambda_2 < 0$: It's a "saddle-node" with one stable eigendirection (corresponding to $\lambda_2 < 0$) and one "center" eigendirection (corresponding to $\lambda_1 = 0$). ps: 如果 $\lambda > 0$ 或都 = 0, 类似分析.

Linearise Stability: 1. Let $N(t) = N^* + n(t)$, $|n(t)| \ll 1$. or $N_t = N^* + n_t$, $|n_t| \ll 1$.

2. $\frac{dn}{dt} = \frac{dN}{dt} = a \cdot f(N^*) + f'(N^*)n(t)$ or $f(N^*) + f'(N^*)n_t$ b. (delay) 直接代入后 linearise.

Justify bifurcation: 1. Assume $n(t) = e^{\lambda t}$ 2. 代入 $\frac{dn}{dt} = \lambda e^{\lambda t} = f'(N^*)e^{\lambda t}$ 3. (Re), (Im) 得两方程.

4a. 反证 $\Re(\lambda) > 0$, < 0 or **4b.find** T_c . Check $T = 0$ stable/unstable $\Rightarrow$ first time $\Re(\lambda) = 0$

Properties for continuous: Time-Independent: At N^* , $N(t - T) = N(t) = N^*$.

Tangency-Condition: At bifurcation point (x^*, r_c) , $f(x^*, r_c) = f(x^*, r_c) = 0$.

Properties for discrete: eigenvalue: $\lambda = f'(N^*)$.

(for perturbations) $\lambda < -1$: $\uparrow$ oscillation; $-1 < \lambda < 0$: $\downarrow$ oscillation;

$0 < \lambda < 1$: $\downarrow$ monotonic; $\lambda > 1$: $\uparrow$ monotonic.

p-periodic: $N^* = f^p(N^*)$. Every p -periodic N^* with $\lambda = -1$ branches into a $2p$ -periodic N^* .

Sarkovskii's Theorem: If a 3-periodic N^* exists, then p -periodic N^* exists for all $p \in \mathbb{N}^+$.

Max/Min: $f'(N_m) = 0$, $N_{\max} = f(N_m)$, $N_{\min} = f(N_{\max})$. assume unimodal, 2-periodic exists.

Types of Bifurcations: Saddle-Node: $\dot{x} = r \pm x^2$ (Normal Form) $0 \rightarrow S \rightarrow U-S$

Transcritical: $\dot{x} = rx - x^2$ (Normal Form) $S-U \rightarrow U-S$

super-critical Pitchfork: $\dot{x} = rx - x^3$ (Normal Form) $S \rightarrow S-U-S$

sub-critical Pitchfork: $\dot{x} = rx + x^3$ (Normal Form) $U \rightarrow U-S-U$

Hopf: (平衡点从稳定变为不稳定, 产生 limit cycle) **Condition:** $\det(J) > 0$, $\frac{d\text{tr}(J)}{dr} \neq 0$. μ : parameter, not t

or: 找到 μ s.t. $\text{tr}(J) > 0$, $\mu > \mu_c$ and $\text{tr}(J) < 0$, $\mu < \mu_c$. **Period:** $T = \frac{2\pi}{\omega}$, 特征向量: $\lambda = \pm i\omega$

Bifurcation Diagram: x-axis: equilibrium value; y-axis: parameter value. 虚线: unstable; 实线: stable

ODEs

$y' + py = q$: Integrating Factor: $I = e^{\int p dt}$ sol: $I(x)y = \int I(x)q dx$

$y'' - \lambda y = 0$: 1. $\lambda > 0$: $y = c_1 \sinh(\sqrt{\lambda}t) + c_2 \cosh(\sqrt{\lambda}t) = c_1' e^{\sqrt{\lambda}t} + c_2' e^{-\sqrt{\lambda}t}$

2. $\lambda = 0$: $y = c_1 t + c_2$ 3. $\lambda < 0$: $y = c_1 \sin(\sqrt{-\lambda}t) + c_2 \cos(\sqrt{-\lambda}t)$

$ay'' + by' + cy = 0$: 1. $\Delta > 0$: $y = c_1 e^{r_1 t} + c_2 e^{r_2 t}$ $r_{1,2} = \mu \pm vi$

2. $\Delta = 0$: $y = e^{\mu t}(c_1 + c_2 t)$ 3. $\Delta < 0$: $y = e^{\mu t}(c_1 \cos vt + c_2 \sin vt)$

4 Nullclines|Limit Cycle|Trapping Region

Nullclines: For: $u' = f(u, v)$, $v' = g(u, v)$. u -nullcline: $f(u, v) = 0$. v -nullcline: $g(u, v) = 0$.

Slope of Nullclines: Since $df = f_u du + f_v dv = 0$ on u -nullcline,

Slope of u -nullcline: $\frac{dv}{du} = -\frac{f_u}{f_v}$. Similarly, $\frac{du}{dv} = -\frac{g_u}{g_v}$ for v -nullcline.

Trapping Region: 1. 直线/nullcline 围闭合区域. 2. 边界上, 外法向量 $\mathbf{n} \cdot \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix} < 0$

Poincaré-Bendixon: One unstable equilibria (no saddle) + trapping region = stable limit cycle exists.

Useful Tools Polar Coord: $rr' = xx' + yy'$ and $r^2\theta' = xy' - yx'$. $\Delta u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$

5 Diffusion|Wave Model

Diffusion Eq: $u(x, t)$ s.t. $u_t = D\Delta u$ with $u(x, 0) = f(x)$ D : diffusion coeff.

Fundamental sol: $\phi(x, t) = \frac{Q}{(\sqrt{4\pi Dt})^{n/2}} e^{-\frac{|x|^2}{4Dt}}$ If $f(x) = Q\delta(x)$, $u(x, t) = \phi(x, t)$

General sol: $u(x, t) = \int_{\mathbb{R}^n} \phi(x - y, t) f(y) dy$

Non-Homo: If $u_t \rightarrow u_g = D\Delta u + a(t)u$, consider $u(x, t) = v(x, t)e^{\int_0^t a(s)ds}$ with $v_t = D\Delta v$.

Flux (J): 1. 考虑 u , J for continuous 函数. (尤其是分段函数) 2. J 考虑物理意义, 边界上为 0.

e.g. $J = -\mu b_x + \delta b(\ln s)$ μ : diffusion coeff. $\frac{\delta}{\mu}$: chemotactic sensitivity

Interpret Initial val. $u_0\delta$: Assumption that the {species} were released at a single point and nowhere else.

Give initial condition: Usually using $u(x, 0) = u_0$ (uniform distribution)

Give boundary condition: Using u , J is continuous at boundary. $\Rightarrow$ 考虑边缘要求连续得到条件

Wave Model: Consider $u(x, t) = u(x - ct) = U(z)$ with $z = x - ct$ c : wave speed

Combaine diffusion: $u_t = Du_{xx} + f(u) \Rightarrow$ ODE: (换元后得到) $-cU' = DU'' + f(U)$

ODEs: (Let $V = U'$) $\begin{cases} U' = V \\ V' = -\frac{c}{D}V - \frac{f(U)}{D} \end{cases} \Rightarrow \begin{cases} \text{If heteroclinic:} \\ \text{If wavefront: (or)} \end{cases} \begin{cases} U(\pm\infty) = u^\pm \\ (u^\pm, 0) \text{ are equilibria of ODEs} \end{cases}$

Homoclinic: $U(\pm\infty) = u^*$ (i.e. $u^+ = u^-$) **ahead of wave:** u^- **behind wave:** u^+

Condition if wavefront: $\lim_{z \rightarrow \pm\infty} U(z) = u^\pm$ and $\lim_{z \rightarrow \pm\infty} U'(z) = 0$

Show the existence of wavefront|connection|heteroclinic between u^- and u^+ :

1. 确定 $u^\pm$, 题目会给, 或 by $f(u)$ 的零点确定. 2. 计算 Jacobian, 确定 $u^\pm$ 的类型且计算 eigenvalues.

3. (ps: for $J = [0, 1; a, b]$ have eigenvectors: $[1; \lambda]^\top$) The positive|negative eigenvalues

at $u^-|u^+$ corresponding to an eigenvector $[1; \lambda]^\top$, which means that it has an unstable|stable eigendirection with positive|negative slope. 4. Therefore, it's plausible

Wave Speed: 1. For $-cU' = DU'' + f(U)$ 两边乘 U' 对 z 在 $(-\infty, \infty)$ 积分 $= 0$

2. $-c \int_{-\infty}^{\infty} (U')^2 dz = D \int_{-\infty}^{\infty} U'' U' dz + \int_{-\infty}^{\infty} f(U) U' dz$ by $U' dz = dU$, $\int_{-\infty}^{\infty} U'' U' dz = \frac{1}{2} [(U')^2]_{-\infty}^{\infty} = 0$

3. $c = \frac{[\int_{-\infty}^{\infty} f(U) dU]}{[\int_{-\infty}^{\infty} (U')^2 dz]}$ Remark: Sign of c depends on area between $f(U)$ and U -axis

Meaning: critical front speed: $c = 0$ (对于灾害蔓延); $c < 0 < c > 0$: no spread(dies out)|spread(occurs)

Possible $c = 0$: Assume continuous dependence of c on {parameter p }. 证明有 $c > 0 < c < 0 \Rightarrow \exists p_c$ s.t. $c = 0$

Fisher-KPP: $u_t = Du_{xx} + ru(1 - \frac{u}{K}) \Leftrightarrow -cU' = DU'' + ru(1 - \frac{u}{K})$, wavefront s.t. $c \geq c_{\min} = 2\sqrt{rD}$.

Condition for Wavefront sol: eigenvalue|separation by var. 算出来后 NO oscillation sol. (i.e. NO sin/cos)

Remark: 如遇计算 parameter 值 | 计算待定系数 | 算 wave speed sign: 直接积分 or 乘 g , U 分部积分!

6 Lotka-Volterra Model — Predator-Prey Model

Model: $\begin{cases} \dot{x} = x(a - by) \\ \dot{y} = y(c - d) \end{cases} \Rightarrow \begin{cases} x: \text{prey pop.} \\ y: \text{predator pop.} \end{cases} \begin{cases} a: \text{prey growth rate.} \\ c: \text{predator growth rate coeff.} \\ b: \text{predation rate coeff.} \\ d: \text{predator death rate.} \end{cases}$

Constant of Motion: 1. 用 $\frac{dy}{dx} = \frac{y}{x}$ 算出 trajectory 方程 e.g. $H(u, v) = \alpha u + v - \ln(u^\alpha v) = H_0$

2. If $\frac{dH}{dt} = 0$ or $\frac{dH}{dt} = 0$, H is constant of motion.

Show trajectory closed: a. Calculate $\begin{bmatrix} H_u & H_v \\ H_{uu} & H_{vv} \end{bmatrix}$ at equilibrium (u^*, v^*) .

b. If positive|negative definite $(\lambda_i > 0 | \lambda_i < 0)$, then H strictly convex|concave, has global min|max, closed.

c. Therefore, trajectory is closed. **Closed trajectory $\Rightarrow$ limit cycle.** (periodic sol)

Why periodic occurs: As any trajectory must lies on one of the closed level curves of H , and we have shown that it is closed. Thus, the periodic oscillation occurs.

Principal of Competitive Exclusion:

If two species compete for the same limited resource, one will outcompete and exclude the other. ps: 图 b,c,d 都是

Competition Model:

$\begin{cases} \dot{x} = x(r_1 - b_1 x - c_1 y) \\ \dot{y} = y(r_2 - b_2 x - c_2 y) \end{cases} \Rightarrow \begin{cases} x, y: \text{pop. of species 1, 2} \\ b_i: \text{intraspecific competition} \\ c_i: \text{interspecific competition} \end{cases}$

右图描述: (a): Stable coexistence. (Weak competition)

intraspecific competition > interspecific competition

(b): Bistability. (Strong competition, by initial condition 决定)

interspecific competition > intraspecific competition

(c), (d): One species outcompetes the other. interspecific competition > intraspecific competition

(e): Weak mutualism. Both species tend to a stable coexistence, benefiting from each other

(f): Strong mutualism. Due to absence of attractor, populations grow without bound, resulting population explosion.

7 Reaction | Reduced, Layer Problem

Complex: 对于反应 e.g. $S + E \xrightleftharpoons[k_2]{k_1} SE \xrightarrow{k_3} P + E$. E : enzyme (酶), SE : complex. 快速出现后稳定, P : 产物

Law of Mass Action: e.g. For reaction $\alpha_1 A + \alpha_2 B \xrightleftharpoons[k_2]{k_1} \beta_1 C + \beta_2 D$ e.g. $[C] = b[A]^{\alpha_1}[B]^{\alpha_2} - a[C]^{\beta_1}[D]^{\beta_2}$

Singular Perturbation Systems: $0 < \varepsilon \ll 1$

fast $\begin{cases} \dot{u} = f(u, v, \varepsilon) \\ \dot{v} = g(u, v, \varepsilon) \end{cases} \xrightarrow{\varepsilon \rightarrow 0} \text{slow} \begin{cases} \varepsilon u' = f(u, v, \varepsilon) \\ v' = g(u, v, \varepsilon) \end{cases} \xrightarrow{\varepsilon \rightarrow 0} \text{layer} \begin{cases} \dot{u} = f(u, v, 0) \\ \dot{v} = g(u, v, 0) \end{cases} \xrightarrow{\varepsilon \rightarrow 0} \text{reduced} \begin{cases} 0 = f(u, v, 0) \\ 0 = g(u, v, 0) \end{cases}$

Inner sol: sol of fast time (layer prob.) **Outer sol:** sol of slow time (reduced prob.) | u : fast var.; v : slow var.

Quasi-Steady State Approximation (QSSA): Assume slow system $\varepsilon \rightarrow 0$ (reduced), find outer sol.

Solve inner|outer sol $\rightarrow$ reduced|layer prob.: (还可直接计算, 对于 manifold 类型的)

1. Assume: **Inner sol** $\begin{cases} U(t; \varepsilon) = u(\tau; \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n U_n(t) \\ V(t; \varepsilon) = v(\tau; \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n V_n(t) \end{cases} \xrightarrow{\varepsilon \rightarrow 0} \text{Outer sol} \begin{cases} u(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n u_n(\tau) \\ v(\tau, \varepsilon) = \sum_{n=0}^{\infty}$

coalesce : 两点合并	eradicated: 根除	membrane potential: 膜电位	unimodal: 单峰	vaccination: 疫苗接种
predator: 捕食者	prey: 被捕食者	trnsriptase: 反转录酶	invades: 入侵	plausible: 似乎合理的
invasion: 入侵	intraspecific competition: 种内竞争	interspecific competition: 种间竞争		
promote: 促进	suppress: 抑制			
boundary region = trapping region = invariant region = confined region				